Notes: Mian, Sufi, and Trebbi (JF, 2015)
Foreclosures, House Prices, and the Real Economy

Contents

1 Big picture

- **Question.** Did the 2007–2009 foreclosure wave depress house prices and real activity, or were foreclosures merely a symptom of falling prices and weak demand?
- **Core shock.** Judicial foreclosure requirements slow the conversion of delinquency into foreclosure; nonjudicial states allow faster foreclosure.
- **Main result.** Nonjudicial states had more than twice the foreclosure rate of judicial states during 2008–2009, and this higher foreclosure intensity caused large house-price and real-activity declines.
- **Economic takeaway.** Foreclosures were not just a consequence of the housing bust; they amplified the bust through fire-sale pressure.

2 Incremental contribution of this paper

- It uses state foreclosure law as an instrument to identify the causal effect of foreclosures on house prices and real activity.
- It separates delinquency/default from foreclosure, showing similar delinquency rates but sharply different foreclosure rates.
- It adds zip-code border evidence showing discontinuities in foreclosure propensity at legal borders but not in precrisis fundamentals.
- It studies the full foreclosure cycle, including the recovery when foreclosure differences disappear.
- It quantifies aggregate importance: about one-third of house-price declines and around one-fifth of residential-investment and auto-sales declines.

3 Data and measurement

3.1 Foreclosures and delinquencies

Foreclosures are measured using RealtyTrac filings. Delinquencies are measured from Equifax as mortgage accounts at least 60 days past due. Outcomes are scaled by homeowners, delinquent accounts, or state/zip-code populations as appropriate.

Interpretation. The design separates borrower distress from legal conversion of distress into foreclosure.

3.2 House prices and real activity

House prices come from Zillow and FCSW/CoreLogic-style indexes. Residential investment is measured using residential permits. Durable consumption is proxied by auto sales from R. L. Polk.

Interpretation. These outcomes test whether foreclosure fire sales spill over to the real economy.

4 Models and empirical specifications

4.1 First stage

$$Foreclosures_s = \alpha + \pi JudicialRequirement_s + \Gamma Controls_s + \varepsilon_s.$$

Interpretation. $\pi < 0$ means judicial requirements reduce foreclosure intensity.

4.2 Second stage

$$Outcome_s = \alpha + \beta \widehat{Foreclosures}_s + \Gamma Controls_s + u_s.$$

Interpretation. β measures the causal effect of legal-induced foreclosure intensity on house prices or real activity.

4.3 Border specification

$$Y_{zbs} = \alpha_b + \gamma Nonjudicial_s + f(distance_{zbs}) + X'_{zbs}\beta + \eta_{zbs}.$$

Interpretation. Border fixed effects compare nearby zip codes on different sides of state legal borders.

5 Identification and empirical design

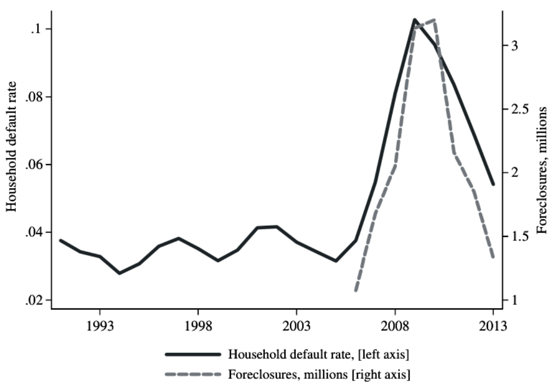
The core exclusion restriction is that judicial foreclosure requirements affect outcomes through foreclosure timing and intensity, not through unrelated state-level shocks. The paper supports this with covariate balance tests, border discontinuities, placebo tests, alternative law controls, and recovery-period evidence.

6 Tables and figures

6.1 Figure 1: Household default rate and foreclosures + Table I: Summary Statistics

Read: Figure 1 shows the national foreclosure wave. Table I summarizes the state-level crisis variables.

Economic message. The foreclosure wave was large enough to plausibly amplify the housing bust.



(a) Figure 1: Household default rate and foreclosures

Table I

Summary Statistics

This table presents summary statistics for the state-level data used in the analysis. Foreclosures are measured by RealtyTrac.com as new foreclosure filings. Delinquencies represent the number of delinquent accounts 60 days past due as measured by Equifax. The scalar homeowner represents the number of mortgage accounts as of 2005 as measured by Equifax. Subprime consumer fraction is the fraction of consumers with a credit score less than 660 as measured by Equifax. Residential permits represent the value of permits for new residential construction as measured by the Census. Auto sales are measured by R.L. Polk.

	N	Mean	*SD*	10^{th}	50^{th}	90^{th}
Foreclosures per homeowner, 2008 and 2009	51	0.028	0.027	0.006	0.020	0.055
Delinquencies per homeowner, 2008 and 2009	51	0.095	0.042	0.058	0.086	0.133
Zillow house price growth, 2002 to 2006	45	0.326	0.163	0.133	0.330	0.588
Zillow house price growth, 2006 to 2007	47	−0.018	0.047	−0.083	−0.014	0.041
Zillow house price growth, 2007 to 2009	48	−0.119	0.126	−0.268	−0.091	0.012
FCSW house price growth, 2002 to 2006	24	0.364	0.199	0.094	0.347	0.674
FCSW house price growth, 2006 to 2007	24	−0.070	0.069	−0.194	−0.049	−0.002
FCSW house price growth, 2007 to 2009	24	−0.205	0.162	−0.475	−0.177	−0.065
Residential permits growth, 2002 to 2006	51	0.289	0.275	−0.071	0.245	0.656
Residential permits growth, 2006 to 2007	51	−0.198	0.141	−0.339	−0.191	−0.037
Residential permits growth, 2007 to 2009	51	−0.768	0.270	−1.082	−0.726	−0.496
Auto sales growth, 2004 to 2006	51	−0.020	0.123	−0.116	−0.046	0.093
Auto sales growth, 2006 to 2007	51	−0.022	0.056	−0.104	−0.019	0.050
Auto sales growth, 2007 to 2009	51	−0.413	0.157	−0.578	−0.399	−0.238
New mortgages/income, 2005	51	0.154	0.077	0.088	0.133	0.227
Debt to income increase, 2002 to 2005	51	0.267	0.156	0.103	0.248	0.463
Subprime consumer fraction, 2000	51	0.329	0.066	0.264	0.305	0.434
Ln(Income, 2005)	51	3.933	0.163	3.740	3.908	4.155
Fraction with income less than 25K, 2005	51	0.434	0.041	0.379	0.432	0.486
Unemployment rate, 2000	51	0.058	0.014	0.044	0.057	0.075
Poverty fraction, 2000	51	0.122	0.032	0.090	0.114	0.164
Black fraction, 2000	51	0.100	0.110	0.005	0.067	0.263
Hispanic fraction, 2000	51	0.055	0.068	0.009	0.030	0.127
Less than high school education fraction, 2000	51	0.182	0.043	0.129	0.180	0.247
Urban fraction, 2000	51	0.713	0.156	0.519	0.714	0.908

(b) Table I: Summary Statistics

6.2 Figure 2: States with judicial foreclosure requirement + Figure 3: Effect on actual foreclosures

Read: Figure 2 maps legal variation. Figure 3 shows the first-stage foreclosure difference.

Economic message. The law changes the speed of foreclosure, not the initial delinquency shock.

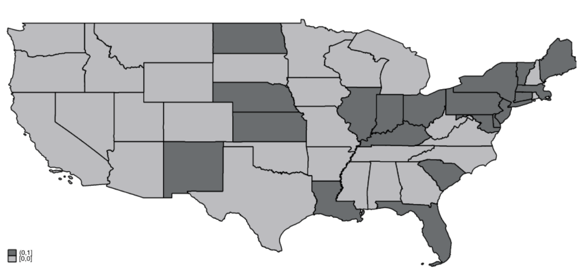


Figure 2. States with judicial foreclosure requirement. States shaded in dark gr

(a) Figure 2

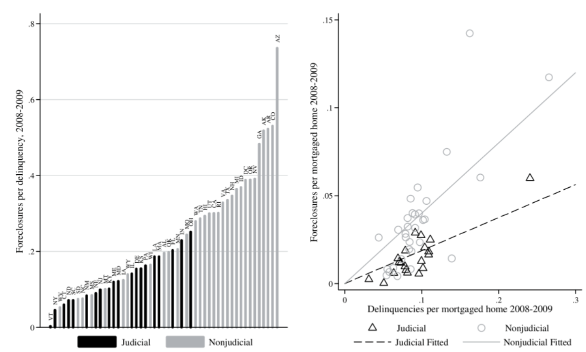


Figure 3. The effect of judicial foreclosure requirements on actual foreclosures. The left panel plots the foreclosure per delinquent account ratio for 2008 and 2009 by state. States that require a judicial foreclosure are shown in black. The right panel plots foreclosures against delinquencies, where the sample is split by whether the state requires a judicial foreclosure.

B. Do Foreclosure Laws Effect Foreclosure Propensity?

(b) Figure 3

6.3 Table II: Judicial Foreclosure Requirement Instrument + Table III: State balance

Read: Table II reports the first stage and its timing. Table III shows that judicial and nonjudicial states are similar on precrisis observables.

Economic message. These tables establish instrument relevance and support the exclusion restriction.

Table II
Judicial Foreclosure Requirement Instrument

Panel A presents coefficients from the first stage regression of foreclosures during 2008 and 2009 on whether a state requires a judicial foreclosure. Panel B repeats the first stage regression separately for each year from 2006 through 2013. Standard errors are heteroskedasticity-robust.

Panel A: First Stage

	Foreclosures per Homeowner 2008–2009	Delinquencies per Homeowner 2008–2009	Foreclosures per Homeowner 2008–2009	Foreclosures per Delinquency 2008–2009
	(1)	(2)	(3)	(4)
Judicial foreclosure requirement	−0.020**	−0.004	−0.018**	−0.167**
	(0.006)	(0.012)	(0.004)	(0.032)
Delinquencies per homeowner, 2008–2009			0.458**	
			(0.081)	
Constant	0.036**	0.096**	−0.009	0.295**
	(0.006)	(0.008)	(0.006)	(0.029)
N	51	51	51	51
R^2	0.134	0.003	0.639	0.287

(Continued)

Table II Panel A

Table II—*Continued*

Panel B: First Stage by Year

	Foreclosures per Homeowner in							
	2006	2007	2008	2009	2010	2011	2012	2013
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Judicial foreclosure requirement	−0.004*	−0.005**	−0.007**	−0.012**	−0.011**	−0.010**	−0.005*	−0.001
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Delinquencies per homeowner	0.217*	0.300**	0.249**	0.348**	0.365**	0.248**	0.095*	0.110*
	(0.086)	(0.074)	(0.058)	(0.057)	(0.050)	(0.072)	(0.041)	(0.042)
Constant	0.000	−0.001	−0.000	−0.007+	−0.007*	−0.002	0.005*	0.000
	(0.002)	(0.002)	(0.003)	(0.004)	(0.003)	(0.004)	(0.002)	(0.002)
N	51	51	51	51	51	51	51	51
R^2	0.238	0.322	0.459	0.699	0.678	0.482	0.150	0.224

**, *, and + are coefficients statistically different from zero at the 1%, 5%, and 10% confidence level, respectively.

Table II Panel B

Table III
Are Judicial Foreclosure Law States Different?

Each row of the table represents a univariate regression of the variable in the first column on whether a state requires a judicial foreclosure. Standard errors are heteroskedasticity robust.

	Judicial Foreclosure Requirement	N	R^2
Delinquencies per homeowner, 2006	0.0014	51	0.003
	(0.004)		
Delinquencies per homeowner, 2009	−0.0028	51	0.001
	(0.010)		
Log Zillow house price, 2005	−0.0023	47	0.00
	(0.13)		
Zillow house price growth, 2002 to 2005	−0.001	45	0.000
	(0.051)		
FCSW house price growth, 2002 to 2005	0.049	24	0.018
	(0.073)		
Debt to income increase, 2002 to 2005	−0.026	51	0.007
	(0.042)		
Subprime consumer fraction, 2000	−0.0161	51	0.014
	(0.018)		
Ln(Income, 2005)	0.0332	51	0.010
	(0.050)		
Fraction with income less than 25K, 2005	−0.0046	51	0.003
	(0.012)		
Unemployment rate, 2000	−0.0046	51	0.029
	(0.004)		

(a) Table III

Table IV
Borders of States with Different Foreclosure Rules

Each row of the table represents a univariate regression of the variable in the first column on whether a state requires a judicial foreclosure. Standard errors are heteroskedasticity robust.

		Number of Zip Codes		
	Border	Within 50 miles of border	Within 25 miles of border	Within 10 miles of border
1	AL-FL	182	94	41
2	AR-LA	103	57	28
3	AZ-NM	85	53	12
4	CO-KS	47	27	11
5	CO-NE	68	31	14
6	CO-NM	93	48	12
7	CT-RI	150	82	40
8	DC-MD	215	128	64
9	FL-GA	199	101	30
10	GA-SC	308	170	77
11	IA-IL	353	192	85
12	IA-NE	301	167	83
13	IL-MO	556	345	176
14	IL-WI	371	162	70
15	IN-MI	268	140	50
16	KS-MO	415	252	131
17	KS-OK	269	124	56
18	KY-MO	62	38	14
19	KY-TN	467	198	77
20	KY-VA	239	165	78
21	KY-WV	286	172	66
22	LA-MS	354	139	51
23	LA-TX	234	115	40
24	MA-NH	295	226	88
25	MA-RI	277	201	83
26	MD-VA	487	321	166
27	MD-WV	152	114	70
28	ME-NH	200	124	63
29	MI-OH	337	134	56
30	MN-ND	201	110	50
31	MO-NE	41	25	11
32	MT-ND	53	28	10
33	NC-SC	458	253	115
34	ND-SD	134	64	30
35	NE-SD	171	97	47
36	NE-WY	37	21	5

(b) Table IV

6.4 Figure 4/Figure 5/Table V: Border tests

Read: Figure 4 shows a foreclosure discontinuity at legal borders. Figure 5 and Table V show no comparable discontinuity in fundamentals.

Economic message. The border design strengthens the causal interpretation.

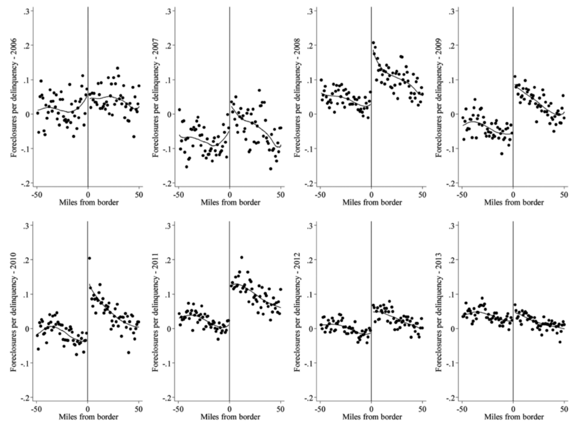


Figure 4. Foreclosures and judicial requirement: Zip codes near the border. These figures plot averages of foreclosures near borders where the judicial requirement regime changes

(a) Figure 4

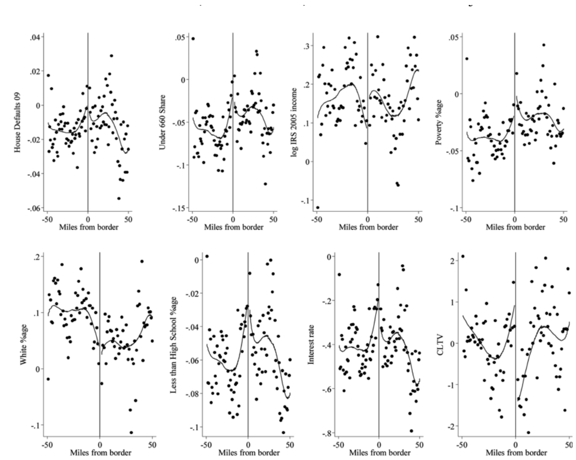


Figure 5. Other variables and judicial requirement: Zip codes near the border. These figures plot averages of variables near borders where the judicial requirement regime change

(b) Figure 5

Table V
Foreclosures at State Border

This table presents tests for discontinuity in foreclosure rate at the state border using zip code-level data. Distance is defined in miles from the border divided by 1,000 and is multiplied by −1 for judicial states.

	Foreclosures per Delinquent Mortgage in							
	2006 (1)	2007 (2)	2008 (3)	2009 (4)	2010 (5)	2011 (6)	2012 (7)	2013 (8)
Judicial foreclosure requirement	0.013	−0.099	−0.190*	−0.170**	−0.195**	−0.146*	−0.077*	−0.036
	(0.067)	(0.070)	(0.078)	(0.047)	(0.051)	(0.058)	(0.036)	(0.033)
Distance	1.490	−0.042	−2.319	−1.990	−3.251*	−1.038	−0.841	−1.410
	(1.855)	(2.635)	(1.860)	(1.780)	(1.501)	(1.515)	(1.231)	(1.073)
Distance squared	−9.369	−28.422	−19.115	−12.649	−14.185	−6.988	−7.737	−3.130
	(20.911)	(32.457)	(16.254)	(14.739)	(17.005)	(10.705)	(10.927)	(9.958)
Distance cubed	−693.472	−749.139	214.943	190.733	591.298	−40.006	−45.630	290.781
	(888.528)	(1,309.71)	(843.873)	(624.268)	(561.586)	(556.379)	(386.404)	(389.672)
State-border × 10-mile strips FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
N	4,918	5,314	5,638	6,036	5,987	5,831	5,734	5,725
R^2	0.395	0.414	0.436	0.482	0.505	0.521	0.666	0.620

**, *, and + are coefficients statistically different from zero at the 1%, 5%, and 10% confidence level, respectively.

Table V

6.5 Figure 6/Table VI: Foreclosures and house prices

Read: Figure 6 shows lower crisis house-price growth and later rebound in nonjudicial states. Table VI reports 2SLS house-price estimates.

Economic message. Foreclosure fire sales lower prices and explain about one-third of the aggregate price decline.

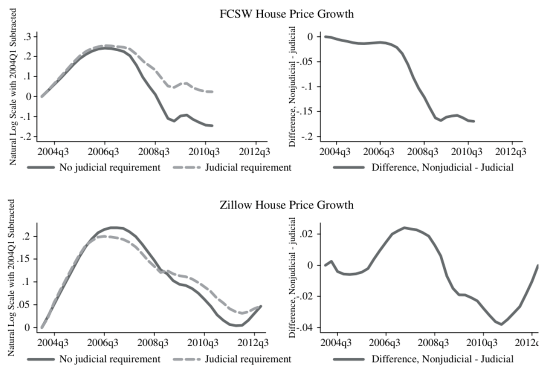


Figure 6. Foreclosures and house prices, reduced form. The figures plot house price growt in judicial and nonjudicial states from 2004 to 2012. Averages are calculated by first calculatir the log difference at time t and 2004Q1 for every state, and then taking the equal-weighted averaş for judicial and nonjudicial categories.

(a) Figure 6

Table VI
Foreclosures and House Prices, State-Level 2SLS

This table presents coefficients of the second stage of a 2SLS specification of house price growth on foreclosures. The first stage, reported in Table II, regresses foreclosures on whether a state has a judicial foreclosure requirement. Standard errors are heteroskedasticity robust.

	Zillow House Price Growth, 2007–2009			FCSW House Price Growth, 2007–2009		
	(1)	(2)	(3)	(4)	(5)	(6)
Foreclosures per homeowner, 2008–2009	−2.245*	−1.751*	−1.966*	−2.277*	−1.579+	−4.089+
	(1.020)	(0.810)	(0.952)	(1.082)	(0.870)	(2.312)
Delinquencies per homeowner, 2008–2009	−1.336*	−0.669	−0.914	−1.497**	−0.308	7.955
	(0.522)	(0.507)	(1.584)	(0.572)	(0.595)	(8.947)
House price growth, 2002–2006		−0.113	−0.200*		−0.105	−0.128
		(0.083)	(0.100)		(0.104)	(0.183)
House price growth, 2006–2007		0.913**	0.407		1.180*	1.934*
		(0.240)	(0.328)		(0.600)	(0.933)
Delinquencies squared, 2008–2009			0.439			−17.544
			(4.552)			(20.878)
New mortgages/income, 2005			0.138			−0.395
			(0.392)			(0.767)
Debt to income increase, 2002–2005			−0.072			−0.074
			(0.102)			(0.167)
Subprime consumer fraction, 2000			−0.175			−0.198
			(0.282)			(0.726)
Income, 2005			0.075			−0.490*
			(0.129)			(0.229)
Income <25K fraction, 2005			−0.088			−3.119*
			(0.598)			(1.534)
Unemployment rate, 2000			−0.087			−8.301**
			(1.673)			(1.862)
Poverty fraction, 2000			1.298			3.364**

(Continued)

(b) Table VI

Table VI—*Continued*

	Zillow House Price Growth, 2007–2009			FCSW House Price Growth, 2007–2009		
	(1)	(2)	(3)	(4)	(5)	(6)
			(0.852)			(1.070)
Black fraction, 2000			0.069			−0.643
			(0.129)			(0.489)
Hispanic fraction, 2000			−0.077			1.040
			(0.189)			(0.892)
<high school education fraction, 2000			−0.016			0.602
			(0.357)			(0.664)
Urban fraction, 2000			−0.144^{+}			−0.138
			(0.085)			(0.242)
Constant	0.072**	0.049	−0.157	0.048	0.009	3.021*
	(0.027)	(0.031)	(0.706)	(0.040)	(0.049)	(1.268)
N	48	45	45	24	24	24
R^2	0.674	0.786	0.849	0.738	0.826	0.909

**, *, and $^{+}$ are coefficients statistically different from zero at the 1%, 5%, and 10% confidence level, respectively.

Table VI continued

6.6 Table VII/Figure 7/Figure 8/Table VIII: Supply and border price effects

Read: Table VII and Figure 7 show higher new listings on the nonjudicial side. Figure 8 shows lower house-price growth near borders. Table VIII documents the recovery.

Economic message. Foreclosure law increases housing supply, depresses local prices, and the effect fades when foreclosure supply dissipates.

Table VII

New For-Sale Listings

This table presents evidence on the effect of foreclosures on new houses listed for sale. Column (1) presents the reduced-form relation between the judicial foreclosure requirement and new for-sale listings. Columns (2) and (3) present coefficients of the second stage of a 2SLS specification of the number of new for-sale listings on foreclosures. The first stage regresses foreclosures on whether a state has a judicial foreclosure requirement. The right-hand side variables are measured as of the same year as the left-hand side. Columns (4) and (5) repeat the state-border discontinuity test for new listings per homeowner. Standard errors for all zip code-level regressions are clustered at the state-border level (40 clusters in total).

	New For-Sale Listings per Homeowner in				
	2009 and 2010 (1)	2009 and 2010 (2)	2009 and 2010 (3)	2009 (4)	2010 (5)
Judicial foreclosure requirement	−0.013+			−0.019**	−0.016**
	(0.007)			(0.004)	(0.005)
Foreclosures per homeowner		0.533*	0.520*		
		(0.222)	(0.231)		
Delinquencies per homeowner			0.112		
			(0.207)		
Constant	0.116**	0.093**	0.085**		
	(0.004)	(0.007)	(0.011)		
Distance				−0.132	−0.192
				(0.181)	(0.223)
Distance squared				−2.903	−1.306
				(2.973)	(2.677)
Distance cubed				27.679	−1.871
				(109.898)	(120.909)
State-border *10-mile strips FE				Yes	Yes
N	51	51	51	8,235	8,235
R^2	0.063	0.386	0.415	0.369	0.335

**, *, and + are coefficients statistically different from zero at the 1%, 5%, and 10% confidence

(a) Table VII

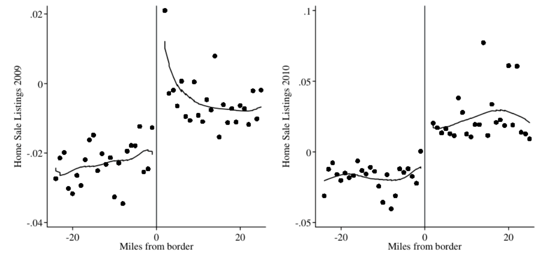


Figure 7. New for-sale listings: Zip codes near border. The figures plot the number of houses newly listed for sale per homeowner in 2009 and 2010 for zip codes that are near borders where the judicial requirement regime changes across states. We generate the graphs by regressing the outcome variable on state-border-group fixed effects and on one-mile band distance-to-the-border dummies (where the dummies assume negative values for judicial states) and then plot the coefficients on the dummies. The border is at zero, the omitted category.

(b) Figure 7

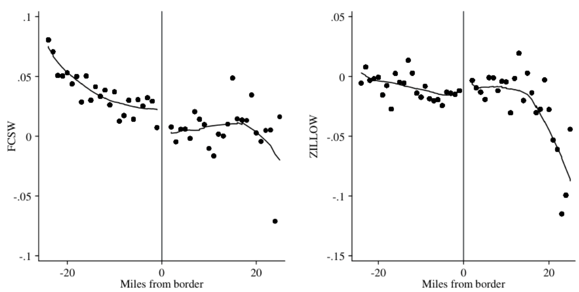


Figure 8. House price growth 2008–2009: Zip codes near border. The figures plot house price growth from 2008 to 2009 for zip codes that are near borders where the judicial requirement regime changes across states. We generate the graphs by regressing the outcome variable on state-border-group fixed effects and on one-mile band distance-to-the-border dummies (where the

(a) Figure 8

Table VIII

House Price Growth, 2007–2012 Judicial versus Nonjudicial States

This table presents coefficients of the reduced-form relation between house price growth and whether a state has a judicial foreclosure requirement. Panel A uses house prices from Zillow, and Panel B uses house prices from CoreLogic. The unit of observation is a state. Standard errors are heteroskedasticity robust.

	Panel A: Zillow			
	House Price Growth 2007–2009 (1)	House Price Growth 2009–2010 (2)	House Price Growth 2010–2012 (3)	House Price Growth 2007–2012 (4)
Judicial foreclosure requirement	0.037*	−0.005	−0.017	0.015
	(0.018)	(0.016)	(0.023)	(0.031)
Delinquencies per homeowner, 2008–2009	−1.444**	−0.350	0.156	−1.638**
	(0.367)	(0.216)	(0.353)	(0.542)
House price growth, 2002–2006	−0.112	−0.000	0.003	−0.109
	(0.086)	(0.048)	(0.082)	(0.097)
House price growth, 2006–2007	1.017**	−0.028	−0.208	0.781+
	(0.252)	(0.209)	(0.316)	(0.392)
Constant	0.059+	−0.026	−0.017	0.015
	(0.033)	(0.019)	(0.023)	(0.031)
N	45	45	45	45
R^2	0.791	0.081	0.054	0.572

(Continued)

(b) Table VIII

Table VIII—*Continued*

Panel B: CoreLogic				
	House Price Growth 2007–2009 (1)	House Price Growth 2009–2010 (2)	House Price Growth 2010–2012 (3)	House Price Growth 2007–2012 (4)
Judicial foreclosure requirement	0.034*	0.011	−0.030*	0.015
	(0.014)	(0.008)	(0.013)	(0.022)
Delinquencies per homeowner, 2008–2009	−0.755*	−0.198	−0.052	−1.005*
	(0.288)	(0.147)	(0.221)	(0.400)
House price growth, 2002–2006	−0.270**	−0.034	0.054	−0.249**
	(0.052)	(0.035)	(0.044)	(0.063)
House price growth, 2006–2007	0.901**	−0.002	−0.200	0.698**
	(0.270)	(0.107)	(0.218)	(0.255)
Constant	0.032	−0.008	0.024	0.048
	(0.028)	(0.010)	(0.023)	(0.034)
N	51	51	51	51
R^2	0.804	0.154	0.187	0.619

**, *, and + are coefficients statistically different from zero at the 1%, 5%, and 10% confidence level, respectively.

Table VIII continued

6.7 Figure 9/Table IX/Table X: Real activity and recovery

Read: Figure 9 and Table IX show effects on residential permits and auto sales. Table X examines the recovery.

Economic message. Foreclosures transmit to construction and durable consumption, explaining roughly one-fifth of those declines.

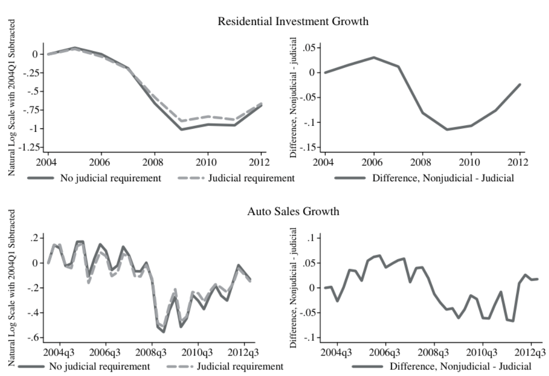


Figure 9. Foreclosures, residential investment, and durable consumption, reduced form. The figures plot residential investment (top) and auto sales (bottom) growth in judicial and nonjudicial states from 2004 to 2012. Averages are calculated by first calculating the log difference at time t and 2004Q1 for every state, and then taking the equal-weighted average for judicial and nonjudicial categories.

(a) Figure 9

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Table IX
Foreclosures, Residential Investment, and Auto Sales 2SLS Estimates

This table presents coefficients of the second stage of a 2SLS specification of residential investment and auto sales growth on foreclosures. The first stage regresses foreclosures on whether a state has a judicial foreclosure requirement. Standard errors are heteroskedasticity robust.

	Residential Permits Growth 2007 to 2009			Auto Sales Growth 2007 to 2009		
	(1)	(2)	(3)	(4)	(5)	(6)
Foreclosures per homeowner, 2008–2009	−6.907*	−6.450*	−2.939	−3.437+	−4.001*	−4.570+
	(3.258)	(3.024)	(3.105)	(1.871)	(2.021)	(2.372)
Delinquencies per homeowner, 2008–2009	−0.962	−1.300	−10.975**	−0.712	−0.174	−2.768
	(1.642)	(1.419)	(3.431)	(0.916)	(0.889)	(3.717)
Growth in LHS variable, 2002–2006		−0.072	−0.215		0.194	0.511**
		(0.104)	(0.195)		(0.194)	(0.119)
Growth in LHS variable, 2006–2007		−0.098	−0.171		0.614	0.453
		(0.202)	(0.226)		(0.495)	(0.403)
Delinquencies squared, 2008–2009			26.969**			11.465
			(10.440)			(9.867)
New mortgages/income, 2005			−0.327			−0.471
			(0.969)			(0.763)
Debt to income increase, 2002–2005			−0.105			0.401
			(0.328)			(0.258)
Subprime consumer fraction, 2000			−0.527			−0.768
			(1.218)			(0.812)
Income, 2005			−0.370			−0.182
			(0.477)			(0.267)
Income <25K fraction, 2005			−1.501			−0.521
			(2.744)			(1.590)

(Continued)

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(b) Table IX

Table IX—*Continued*

	Residential Permits Growth 2007 to 2009			Auto Sales Growth 2007 to 2009		
	(1)	(2)	(3)	(4)	(5)	(6)
Unemployment rate, 2000			-7.088^{+}			0.730
			(3.893)			(1.960)
Poverty fraction, 2000			4.368			0.501
			(3.158)			(1.501)
Black fraction, 2000			0.889			0.397
			(0.554)			(0.334)
Hispanic fraction, 2000			0.268			−0.114
			(0.668)			(0.420)
<high school education fraction, 2000			−0.387			0.376
			(1.419)			(0.921)
Urban fraction, 2000			0.431^{+}			0.123
			(0.225)			(0.207)
Constant	-0.485^{**}	-0.464^{**}	1.990	-0.250^{**}	-0.268^{**}	0.742
	(0.087)	(0.079)	(2.911)	(0.048)	(0.046)	(1.613)
N	51	51	51	51	51	51
R^2	0.398	0.422	0.596	0.356	0.384	0.524

**, *, and $^{+}$ are coefficients statistically different from zero at the 1%, 5%, and 10% confidence level, respectively.

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Table IX continued

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Table X

Residential Investment and Auto Sales Growth, 2007–2012 Judicial versus Nonjudicial States

This table presents coefficients of the reduced-form relation between residential investment growth (Panel A), auto sales growth (Panel B), and whether a state has a judicial foreclosure requirement. The unit of observation is a state. Standard errors are heteroskedasticity robust.

Panel A: Residential Investment

	Residential Permits Growth 2007–2009 (1)	Residential Permits Growth 2009–2010 (2)	Residential Permits Growth 2010–2012 (3)	Residential Permits Growth 2007–2012 (4)
Judicial foreclosure requirement	0.125^{*}	0.017	−0.051	0.091
	(0.061)	(0.031)	(0.073)	(0.114)
Delinquencies per homeowner, 2008–2009	-4.102^{**}	−0.405	0.535	-3.973^{**}
	(0.779)	(0.394)	(1.008)	(1.316)
Residential permits growth, 2002–2006	0.008	−0.105	0.330	0.233
	(0.100)	(0.066)	(0.211)	(0.249)
Residential permits growth, 2006–2007	0.017	−0.110	−0.402	−0.496
	(0.227)	(0.124)	(0.322)	(0.403)
Constant	-0.428^{**}	0.084^{+}	0.074	-0.270^{+}
	(0.100)	(0.044)	(0.088)	(0.146)
N	51	51	51	51
R^2	0.474	0.119	0.194	0.172

(Continued)

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(a) Table X

Table X—*Continued*

Panel B: Auto Sales

	Auto Sales Growth 2007–2009 (1)	Auto Sales Growth 2009–2010 (2)	Auto Sales Growth 2010–2012 (3)	Auto Sales Growth 2007–2012 (4)
Judicial foreclosure requirement	0.074^{*}	0.018	-0.039^{+}	0.053
	(0.037)	(0.028)	(0.021)	(0.044)
Delinquencies per homeowner, 2008–2009	-1.990^{**}	−0.157	0.062	-2.085^{**}
	(0.497)	(0.385)	(0.282)	(0.728)
Auto sales growth, 2002–2006	0.671	0.190	−0.001	0.860^{+}
	(0.444)	(0.430)	(0.285)	(0.463)
Auto sales growth, 2006–2007	0.226	0.436^{+}	−0.074	0.588
	(0.229)	(0.217)	(0.100)	(0.353)
Constant	-0.234^{**}	0.107^{**}	0.190^{**}	0.063
	(0.049)	(0.035)	(0.027)	(0.076)
N	51	51	51	51
R^2	0.496	0.275	0.068	0.448

**, *, and $^{+}$ are coefficients statistically different from zero at the 1%, 5%, and 10% confidence level, respectively.

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(b) Table X continued

7 Short summary

- State foreclosure laws create plausibly exogenous variation in foreclosure intensity.
- Nonjudicial states foreclose much faster, despite similar delinquency rates.
- Foreclosures depress house prices, residential investment, and auto sales.
- The recovery period supports the temporary fire-sale interpretation.

8 Conclusion

8.1 Core conclusion

Foreclosures were a causal amplification mechanism in the Great Recession.

8.2 Incremental lesson

The institutional details of foreclosure law matter for macro outcomes. Legal speed determines how default translates into forced sales, price pressure, and real activity.

8.3 Policy implications

Foreclosure policy affects aggregate demand and housing-market stability, not just private lender-borrower rights. Slower foreclosure may reduce fire-sale externalities, but it may also delay resolution.

8.4 Takeaway

The paper turns foreclosure from a private contractual outcome into a macro-financial amplification channel.